MAJOR PROJECT

Design and Analysis of 4 Channel P-Doped Double Recessed GateAlGaN/GaN based MOS-HEMT

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CERTIFICATE

This is to certify that the project work entitled **Design and Analysis of 4 Channel P-Doped Double Recessed Gate AlGaN/GaN based MOS-HEMT** shows a work done by Mohammad Farukh Zafar and Fahad Zama, in partial fulfillment of requirement for the award of the degree of Bachelor of Technology in Electronics and Communications Engineering at Jamia Millia Islamia, New Delhi, under my supervision and guidance.

I hereby approve it for submission to the university for the award of Bachelor of Technology in Electronics and Communications Engineering. Further, to the best of my knowledge, this report has not been submitted to any other institution for the award of any degree or diploma.

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DECLAERATION

I declare that this dissertation work titled **"Design and Analysis of 4 Channel P-Doped Double Recessed GateAlGaN/GaN based MOS-HEMT"** submitted in partial fulfilments of the degree of B.Tech in (Electronics and Communication Engineering) is a record of original work carried out by me under the supervision of Dr. Md. Waseem Akram, and has not formed the basis for the award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

The physical properties of gallium nitride (GaN) and the related AlGaN/GaN heterojunctions are ideally suited for the fabrication of power semiconductor devices. The high electrical breakdown field strength of GaN in combination with high conductivity due to a two dimensional electron gas by induced polarization enables the development of transistors with high off-state voltages, low on-state resistances and low switching charges. While material quality and process technology are continuously improved, also progress in layout developments is required with the same degree of care.

This work wants to push the GaN-technology by new approaches in design and characterization of highly-efficient GaN transistors in order to release its full potential. Particular emphasis is placed on the development of designs of GaN HEMTs with postive Vth. We have provided a detailed discussion of development of the device using SILVACO TCAD. In this dissertation work, we studied 4 Channel Double Recessed Gate AlGaN/GaN based MOS-HEMT device which can be used in High Power and High Frequency applications.

We have discussed about various materials for the device, transfer characteristics, output characteristics, transconductance curve and various device parameters like Conduction Band Energy, Electric Field, Electron Concentration.

This work contributes substantially to the progress and development of highly-efficient high performance transistors for power applications.

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1 Introduction

1.1 Motivation

Global concerns such as climate change, overpopulation, and limited resources compel us to make a more sustainable use of the precious energy sources. Advanced ways of savings in the fields of energy generation, conversion, distribution, and consumption are required. In this context electric power is one of the most economical and versatile form of energy and power electronics is its enabling technology. Therefore the demand for advanced power electronic systems is continuously growing. The Internet of Things (IoT) is a modern concept that theoretically allows for interaction between most physical items on Earth, including machinery, buildings, even animals, through embedded sensors or chips. Since the turn of the twenty-first century, this area of quickly expanding technology has increased the convenience of daily life for people.

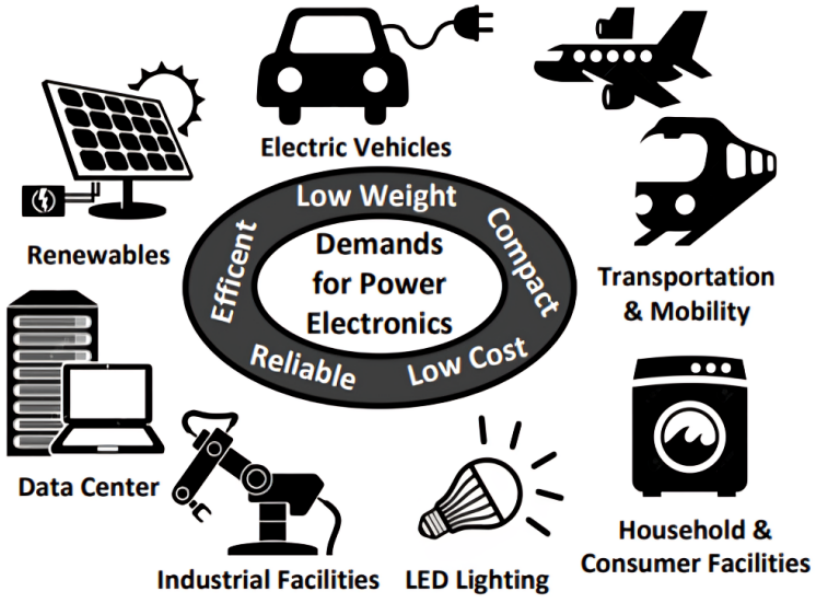


Figure 1: Common demands for prospective power electronics in various applications.

The desires of power electronic systems are recurring. The next generation of power systems has to be high efficient, high compact, with low weight and with low cost. Such developmentsare only conceivable by new high performance power technologies based on wide bandgap semiconductors, such as silicon carbide (SiC) or gallium nitride (GaN), whereas the conventional technologies based on silicon (Si) are already highly developed and the physical capability is almost exhausted.

The development of semiconductor technology allowed the development of a huge industry and a multitude of applications that revolutionized our daily lives. Traditionally, when one mentions of semiconductors, one thinks of silicon, which became the main work horse of modern electron-

ics. Silicon-based transistors have found application in a vast range of domains, but III-V semiconductor technologies are maturing and provide enough performance benefits to justify their adoption in specialized functions. GaAs-based amplifiers are widely used in cellphones and RF applications. InP-based high electron mobility transistors (HEMTs) and heterojunction bipolar transistors (HBTs) are the leading technologies where ultra-high cutoff frequency devices are needed (fT> 300 GHz). In wireless communication, where high-power is required, GaAs transistors and Si-based laterally diffused transistors (LDMOS) have been challenged by the advancement of GaN-based technology. SiC based devices offer an alternative to Si-based transistors in applications where efficient heat dissipation and high breakdown voltage are required, thanks to the better thermal conductivity and breakdown field of SiC. The performance of SiC-based devices ultimately falls short at high frequencies because of its less than optimal transport properties, however this is not the case with GaN.

The main characteristics of the mentioned semiconductors are summarized in Table 1. GaN has the widest bandgap along with the highest critical breakdown field, and a good thermal conductivity that matches silicon. The wide bandgap allows operation at very high temperatures. Therefore, GaN promises to deliver rugged devices which would be able to withstand

Parameter	Si	GaAs	InP	SiC	GaN
E_g (eV)	1.1	1.42	1.35	3.25	3.40
μ $(cm^2/V \cdot s)$	1500	8500	5400	700	1000-2000
$v_{sat}(10^7 cm/s)$	1	1.3	1	2	2.5
$E_{br}(\times 10^6 V/cm)$	0.3	0.4	0.5	3	3.3
κ $(W/cm \cdot K)$	1.5	0.5	0.7	4.5	>1.5

Table 1: Fundamental parameters of the most widely used semiconductors

high voltages and dissipate heat efficiently.

The superior breakdown field and saturated velocity make GaN-based devices capable of delivering higher RF power than GaAs and LDMOS based technologies. The high mobility of GaN and its high critical field allows aggressive scaling of the transistor size, which makes it the leading technology for high-power applications at high frequency. While AlGaN/GaN HEMTs have already found their way into some niche applications, intense research efforts are being devoted to the AlN/GaN, AlInN/GaN and AlInGaN/GaN material systems, which enable higher carrier densities and allow aggressive gate length downscaling to reach even higher frequencies of operation.

1.2 Literature Review

1. In the late 1990s and early 2000s, **AlGaN/GaN HEMT** research was focused on the material growth techniques, device fabrication processes, and **fundamental device properties**. The studies highlighted the advantages of AlGaN/GaN HEMTs, such as high electron mobility, excellent thermal **stability** and potential for high-power and high-frequency applications

2. **Between 2000 and 2006**, researchers continued to explore improvements **in materials** and device **performance**. **Research has** included buffer layer **design**, gate dielectric optimization, and surface passivation to **reduce electron capture** and improve device reliability.

3. **Between 2011 and 2015**, research efforts were concentrated on improving **high frequency** characteristics, **increasing power perfor-**

mance, and investigating novel device architectures for cutting-edge applications. Numerous research emphasized the influence of various material properties on the effectiveness and dependability of devices.

4. **Between 2016 and 2021**, AlGaN/GaN HEMTs have recently been the subject of study concerning their **integration** into various electronic and power applications

5. **Between 2021 and 2023**, research on AlGaN/GaN based HEMT is focused more on **Dimension reduction**, High frequency, **improving Threshold voltage**, High Bandgap Material, No. of Channels and Noise Margin of the device.

6. In the mid-2000s, researchers began to explore the potential benefits of adding a gate oxide layer to AlGaN/GaN HEMTs to form MOS HEMTs. During this time, preliminary research was conducted investigating gate dielectric materials, device fabrication techniques, and basic MOS-HEMT properties.

7. **Between 2011 and 2015**, research efforts in the field of AlGaN/GaN MOS HEMTs intensified, focusing on improving **gate dielectric properties**, gate stack design and achieving higher device efficiency. Issues related to interface traps and gate leakage were also addressed in the studies.

8. **Between 2021 and 2023**, research on AlGaN/GaN based MOS HEMT is focused more on Dimension reduction, **High frequency**, **improving Threshold voltage**, High Bandgap Material, **No. of Channels**

and **Noise Margin** of the device.

1.3 Developments and Challenges for AlGaN/GaN HEMT

The AlGaN/GaN based HEMTs development has started in the 1990s. In the early 1990s, a study shows, how to grow the GaN thin film by molecular-beam epitaxy (MBE) and chemical vapour deposition (CVD) techniques [6], and it allowed the technologies about the growth of bulk GaN and the AlGaN/GaN heterojunction. Khan et al.[7] reported the first HEMT devicewith AlGaN/GaN heterojunction in 1993 and it was grown on the sapphire. At that time, with a large dislocation density in the GaN epitaxy and a poor substrate thermal conductivity, resulted in not suitable for real applications. Then in 1997, the first GaN-on-SiC HEMT is developed by Binary et al. [8]to improve the device performance. The AlN nucleation layer was deposited above the SiC, which reduces the lattice mismatch between the substrate and the GaN layer, therefore, it successfully mitigates the dislocation density in GaN. In the meantime, the SiC substrate with a high thermal conductivity helps to improve the thermal dissipation rate. However, with a relatively high cost for the SiC substrate, the development of GaN HEMTs is mainly pushed by the defence industry. In 2001, Chumbes et al[9]developed the first GaN-on-Si HEMT and a thin GaN layer was adopted to prevent the substrate cracking caused by the different thermal coefficients and the lattice mismatch. Nowadays, the AlN nucleation layer and the AlGaN strain relief layer both are used combined to make the GaN layer strain-free [10].

Other studies focused on device application and reliability exist despite

advancements in device fabrication. Khan et al. [11] has developed the enhanced-mode (E-mode) GaN based HEMTs, also known as the normally-off GaN HEMTs, with a thin AlGaN barrier (10 nm) in 1995. The major advantage of the E-mode device over the depletion-mode (D-mode) device is that it allows the amplifier circuit with a single-polarity voltage supply [12]. The E-mode GaN HEMTs are also achieved by the recessed gate[13], Fluorine implantation [14]and the p gate[15]. Carbon is doped inside GaN buffer layer to reduce the vertical leakage in power ofGaN devices [16], [17]. However, carbon in the GaN buffer layer also induces the buffer charge trapping [18], which results to the dynamic RON increase, so it is a trade-off for the vertical leakage and the dynamic RON with a C-doped GaN buffer layer. Iron(Fe) is also a very common type of dopant to make the GaN buffer as semi-insulating [19]. For some RF GaN HEMTs, ensuring the linearity of the waveform is much more important than controlling of the leakage, therefore, Iron is a better choice than Carbon, as Iron not lead to the significant dynamic RON increase as Carbon. GaN based HEMTs suffers from the surface charge trapping as it acts as a virtual gate to deplete the 2DEG after stress. In 2001, Vetury et al. [20]found that using Silicon Nitride passivation layer above the AlGaN barrier can reduce the surface states in GaN based HEMTs.

To increase the breakdown voltage of the GaN HEMTs, Karmalkar et al. [21] used the field plate to smooth the electric field, avoiding the field peak in the vicinity of the gate edge in 2001.

In 2014, GaN material has become a promising semiconductor since the invention of the blue LED by Akasaki, Amano and Nakamura, and a Nobel

Prize in Physics was also awarded, as it enables white light generation with a high efficiency [22]. In the years that followed, more research on the GaN/HEMT technology was attracted by this exciting news. AlGaN/GaN HEMTs technology has just reached a mature stage of commercialization and has already benefited people all around the world after nearly 30 years of development. According to the Yole Développement market report 2021 [23], With a rapidly expanding market for GaN-based chargers for consumers, as well as for the telecom and automotive sectors, the power GaN market reached approximately $46 million in 2020 and will approach $1.1 billion in 2026.

And a high breakdown voltage [24], of the device with low ON resistance will dissipate less heating energy during the ON state and therefore increase the energy efficiency also. While the ON-resistance mostly depends on the carrier mobility and carrier density, the breakdown field of a semiconductor is connected with the band gap since it dictates the energy necessary for triggering an electron from the valence band to the conduction band. Ga203 is a potential semiconductor with a breakdown field of about 8 MV/cm, although it has a low electron mobility (300 cm^2/Vs) and low thermal conductivity (10 - 30 W/(m·k)) [25][26], makes itnot suitable for high power applications. It is only suitable for some specific applications and is at the early research stage, Diamond shows both the high breakdown field (10 MV/cm) and the high electron mobility (4500 cm^2/Vs) simultaneously [27], Theoretically, this makes it the best material for power applications, but because it lacks n-type dopants and has a high level of resistance, it is currently unsuitable for producing high-quality power devices.

For RF applications like wireless networks, the ideal semiconductor should have great carrier mobility to facilitate quick switching with a low ON resistance. GaAs and InP both exhibit electron mobilities of 8500 cm^2/Vs and 5400 cm^2/Vs that make them appropriate for GHz frequency range operation. They are more susceptible to high voltage because they have more narrow bandgaps than GaN (Eg, GaAs = 1.42 eV, Eg, InP = 1.34 eV). GaN HEMTs are therefore more appropriate for high-power RF applications, such as 5G base stations. Despite recent advancements in AlGaN/GaN HEMT technology, there is still room to improve device performance beyond what the GaN material is capable of. There are many challenges related to the device reliability which need to be solved such as the dynamic RON increase (also known as the current collapse) after device stress [28]. The instability of threshold voltage [29], and the high channel temperature during the high power application [30].

1.4 Thesis Objective

To study the performance of the device when the Gate electrode of the 4 Channel Double Recessed Gate AlGaN/GaN based MOS-HEMT device is Recessed.

1.5 Contribution of Thesis Work

The research identified that an Analog/RF and Power Performance Analysis of an Underlap DG AlGaN/GaN Based High-K Dielectric MOS-HEMT[31] can also work as Normally OFF device i.e. the threshold voltage of up to 1.9V is achieved and the output DC power can also be increased. When

the no of channels is increased in the DC DG MOS-HEMT device and Gate electrode is recessed, the performance of the device can be improved and also Normally OFF device is achievable.

1.6 Thesis Structure

The research work is divided into 5 Chapters. The overview of each chapter is discussed. In Chapter 1, discussed about literature review and basics of heterojunction devices. In Chapter 2, in this Chapter the properties of GaN material and physics behind working of basic HEMT is discussed. In Chapter 3, in this Chapter, about SILVACO TCAD tool is discussed. A DC DG AlGaN/GaN based MOS HEMT based structure is redeveloped and transfer characteristics, output characteristics and various other parameters have been calibrated and discussed. In Chapter 4 we have discussed about Proposed Structure i.e. 4 Channel Double Recessed Gate AlGaN/GaN based MOS-HEMT has been calibrated using SILVACO TCAD tool and Transfer Characteristics, Output Characteristics and various other parameters have been discussed. In Chapter 5, few options have been discussed, using them future researchers can research on for the better operation and performance of the device.

2 GaN Technology

2.1 Gallium Nitride

2.1.1 GaN Properties

Gallium nitride(GaN) is a wide bandgap III-V semiconductor material having a crystal structure of zinc blende or wurtzite. In both crystal structures, the N atoms occupy the tetrahedral vacancies as each N atoms have 4 nearest Ga neighbours in connection. For GaN devices, the wurtzite structure is widely used because it is thermodynamically more stable than zinc blende.

Many useful electronic properties of GaN can be obtained from the energy band structure. Between the top of the valence band and the bottom of the conduction band is where the bandgap is located. The wurtzite GaN has a direct bandgap and the bandgap is 3.44 eV[32].

The carrier mobility μ is an important parameter for semiconductor materials and it is determined by the equation below

$$\mu = e\tau/m$$

where e is the electron charge, τ is the mean free time for the carrier, m is the effective mass. Therefore, the effective carrier mass and band curvature are inversely related to the carrier mass and carrier mobility, respectively. Since they exhibit relatively low mobilities and are often heavier than electrons, holes are not the minority carriers in GaN devices.

Due to impact ionisation, semiconductor materials also degrade in a strong electric field. According to [33], the critical electric field of a direct bandgap

semiconductor has been derived as a function of bandgap

$$E_c = 1.73 \times 10^5 E_g(eV)^5$$

where E_c is the value of the critical electric field and E_g is the value of the bandgap. GaN has a bandgap of 3.44 eV [32], and a critical electric field of 3 MV/cm [5]. It makes GaN suitable for high-power applications.

2.1.2 GaN and other Semiconductors

Each semiconductor material has a distinct property that makes it suitable for a variety of applications. The semiconductor must have a high critical field and a high carrier mobility since the transistor for a power application needs to have a high breakdown voltage and a low ON resistance. Wide bandgap semiconductors have received a lot of attention in the power industry. The properties of major wide bandgap semiconductor materials are shown in Table 2. The semiconductors are usually described with Baliga's figure of merit (BFOM) [34]as follows:

Properties	GaAs	4H-SiC	Si	GaN
E_{gap} [eV]	1.4	3.23	1.12	3.4
$E_{breakdown}$ [MV/cm]	0.4	0.3	2	3
μ_e [cm^2/Vs] 8500	1350	720	900	
Dielectric Constant	12.1	11.8	10	9.5
BFOM	17	1	134	537

Table 2: Comparison of Bulk Semiconductor material properties [5][26][27]

Si is the most prevalent and commonly used semiconductor, however because of its relatively low breakdown field, it is not suitable for high-power applications. GaAs is useful for RF applications because it has better electron mobility than other materials and a low breakdown field. SiC has

established its benefits not only because of its low cost of manufacture but also because of its high heat conductivity. When switching in some power switching circuits, the devices produce a significant quantity of joule heating energy; as a result, they need a high thermal dissipation rate to maintain the desired temperature. In many situations, GaN is a more promising wide bandgap semiconductor than SiC. It's bulk electron mobility is around 900 900cm^2/Vs, the polarization nature of GaN crystal leads to the formation of 2DEG which has a electron mobility up to 2000 cm^2/Vs at room temperature [4]and a high electron density. However, the manufacturing of GaN is more expensive than that of the prior semiconductors, and it typically has a high dislocation density, which makes commercialization difficult.

Ga203 is a possible semiconductor with a breakdown field of about 8 MV/cm for upcoming high-power applications. Limited heat conductivity and limited electron mobility are two drawbacks of Ga203. As a result, it is still in the early stages of research and only appropriate for particular specific applications. Theoretically, diamond materials are the best material for power applications since they simultaneously have a high breakdown field and high mobility. However, since they lack n-type dopants and exhibit a high resistance due to the presence of a 0.3 eV deep acceptor, it is difficult to produce good commercial devices at this time.

For the RF application, the ideal semiconductor should have a high carrier mobility to enable fast switching and low resistance. GaAs and InP (1.34 eV) have exhibited their advantages in the RF application due to their high electron mobility. However, with relatively low bandgaps, GaAs and

InP (5400 cm^2/Vs) are less appropriate for the high-power RF application, such as the power amplifier used for the 5G base station. Thanks to the wide bandgap of GaN and the high electron density within the 2DEG of AlGaN/GaN heterojunction, GaN becomes a promising candidate for high-power RF application.

2.2 Structure of AlGaN/GaN HEMT

A typical AlGaN/GaN HEMT structure is shown in Figure 2. It consists of the substrate, the GaN epitaxy, the AlGaN/GaN heterojunction with 2DEG in between, the passivation layer (SiNx), contacts and field plates.

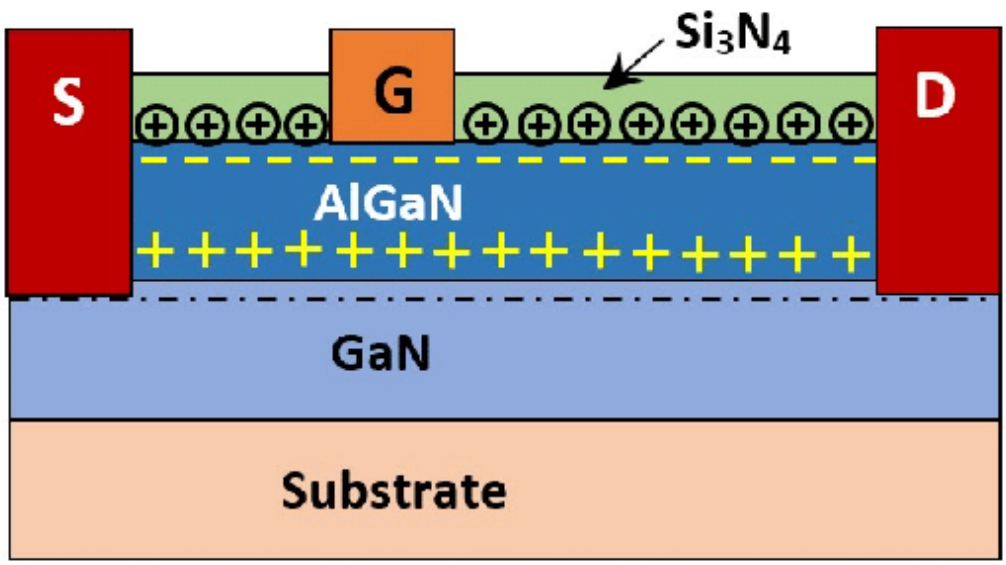


Figure 2: Schematic of a AlGaN/GaN HEMT structure

2.2.1 Substrate

There is multiple substrate choice for the GaN device epitaxy as shown in Table 3. The ideal substrate for GaN should have a high thermal conductivity, a small lattice mismatch with GaN, large wafer size availability and a low cost. Although the bulk GaN can be grown and fits with GaN epitaxy, it is hard to commercialize due to its high cost and small wafer size[35]. The Si wafer is a good substrate choice due to its mature fabrication process, large wafer size and low cost. However, considering that GaN HEMTs are widely used for the r and RF applications and a high thermal conductivity of substrate is often required. Therefore, SiC becomes a better choice for GaN HEMTs not only due to its high thermal conductivity but also the small lattice mismatch with GaN. Meanwhile, the demand of low substrate loss makes SiC better than Si for RF operation [9]. Diamond benefits from its extremely high thermal conductivity and is a potential option for the next generation of GaN HEMTs [30].

	GaN	Si	SiC	Sapphire	Diamond
E_{gap} [eV]	3.4	1.1	3.3	9.9	5.5
Thermal Conductivity [W/cm·K]	2	1.5	4.9	0.35	180
Lattice mismatch with GaN (%)	0	17	3.5	14	89
Size (mm)	30	300	150	150	10
Cost	High	Low	High	Medium	Extremely High

Table 3: Comparison of Bulk Semiconductor material properties [5][26][27]

2.2.2 Nucleation and Strain Relief Layer

For GaN-on-Si devices, the lattice mismatch Si and GaN will result in a large dislocation density and even cracks on the wafer. AlN is a common choice for nucleation followed by the strain relief layer to mitigate the strain due to the lattice mismatch. For GaN-on-SiC devices, AlGaN can be used as the nucleation layer with a small lattice mismatch and then GaN can be grown on the AlGaN layer.

2.2.3 Passivation

The passivation layer is deposited above the AlGaN barrier and the GaN cap to reduce the surface traps and the leakages. Without the passivation, charges will be stored on the AlGaN surface and act as a virtual gate to increase the channel resistance [23].

2.2.4 AlGaN/GaN Heterojunction and 2DEG

The AlGaN barrier is deposited above the GaN channel to form the AlGaN/GaN heterojunction to generate the 2DEG. For wurtzite GaN and AlN, there is spontaneous polarization (PSP). The spontaneous polarization is AlGaN is between that of AlN and GaN. If the AlGaN layer is grown on a bulk GaN, the lattice discontinuity will induce tensile stress within the AlGaN layer. The tensile stress results in thezoelectric polarization (PPZ). The schematic of polarization in AlGaN/GaN heterojunction is shown in Figure 3. Note that the spontaneous polarization of AlGaN is larger than that of GaN, so there will be net positive polarization charge on the AlGaN/GaN interface. The sign of the interface charge determines

the band bending direction, as shown in Figure 2(b). The conduction band of GaN near the interface is pushed down to the Fermi level due to the net positive polarization charge, which gives rise to the 2DEG. However, due to the electron conservation, there should be an electron source providing the free electrons to 2DEG. According to [36], the surface donors of AlGaN take this role. The donors above the Fermi level become ionized and leave a positive charge on the AlGaN surface. It is the main reason for the formation of the 2DEG in the AlGaN/GaN heterojunction.

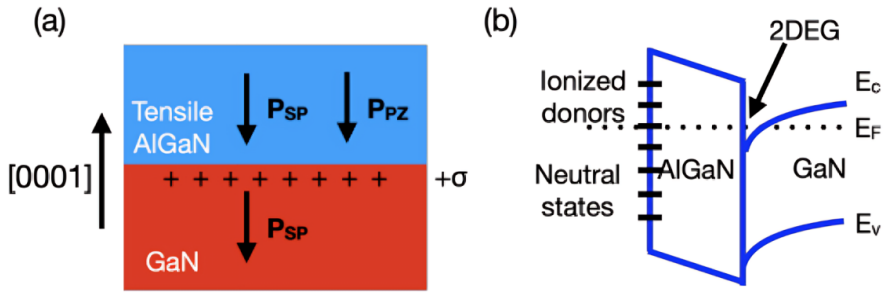


Figure 3: : (a) AlGaN/GaN heterojunction with the spontaneous polarization (PSP) and the piezoelectric polarization (PPZ). (b) Band diagram of the AlGaN/GaN heterojunction with the formation of 2DEG. A part of donors on the AlGaN surface is ionized, proving the electron source for 2DEG. [22]

The electrons in the quantum well are described to form a "two-dimensional electron gas" (2DEG), as indicated by the dashed line in Fig. 3(a). Most electrons are found within a thickness of a few nm, while few electrons concentration decay exponentially towards the buffer. A small fraction is located in the barrier layer. It is highly desired to confine the carrierelectrons to the quantum well. If the electrons are located in the barrier layer, they will have considerably lower mobility, while wide distribution into

the buffer layer indicates that the device may suffer short-channel effects at high electric fields. The confinement of the 2DEG at the channel-barrier interface plays a crucial role in the HEMT performance.

2.2.5 The Basic GaN Transistor Structure

The basic depletion-mode GaN transistor structure is shown in Figure 4. As with any power FET, there are gate, source, and drain electrodes. The source and drain electrodes pierce through the top AlGaN layer to form an ohmic contact with the underlying 2DEG. This creates a short circuit between the source and the drain until the 2DEG of electrons is depleted, and the semi-insulating GaN crystal can block the flow of current.

In order to deplete the 2DEG, a gate electrode is placed on top of the AlGaN layer. When a negative voltage relative to both drain and source electrodes is applied to the gate, the electrons in the 2DEG are depleted out of the device. This type of transistor is called a depletion-mode, or d-mode, HEMT. The basic AlGaN/GaN HEMTs operates in Depletion mode (D-mode) or normally-ON operation, which limit their use in power applications due to safety reasons. Since interface due to intrinsic spontaneous and piezoelectric polarization effects, a negative bias has to be applied at the gate to switch the device OFF.

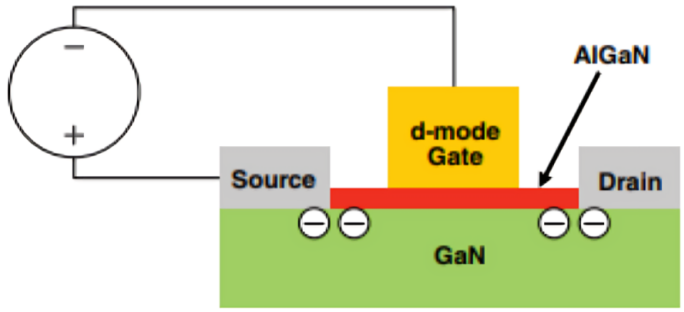


Figure 4: Depletion-mode (d-mode) HEMT[37]

There are two common ways to produce a d-mode HEMT device:

1. A transistor having a Schottky gate electrode that was created by depositing a metal layer directly on top of the AlGaN. The Schottky barrier was formed using metals such as Ni-Au.

2. Depletion-mode devices have also been made using an insulating layer and metal gate similar to a MOSFET, both types are shown in Figure 5.

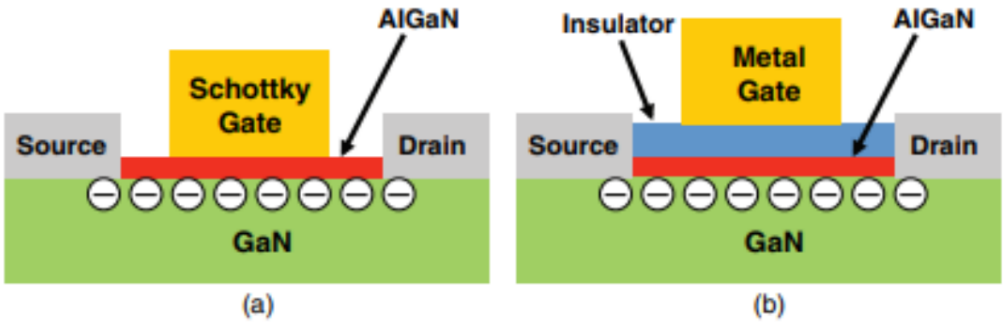


Figure 5: Cross section of a basic depletion-mode GaN HEMT with (a) Schottky gate, or (b) insulating gate[37]

In power applications, depletion mode devices are not convenient because, at the startup of a power converter, a negative bias must first be applied to the power devices. If this negativebias is not applied first, a short circuit will result.

An enhancement mode device, on the other hand, would not suffer from this limitation because an e-mode device is normally-OFF device as shown in Figure 6(a) and will not conduct current until a positive voltage is applied to the gate, as illustrated in Figure 6(b).

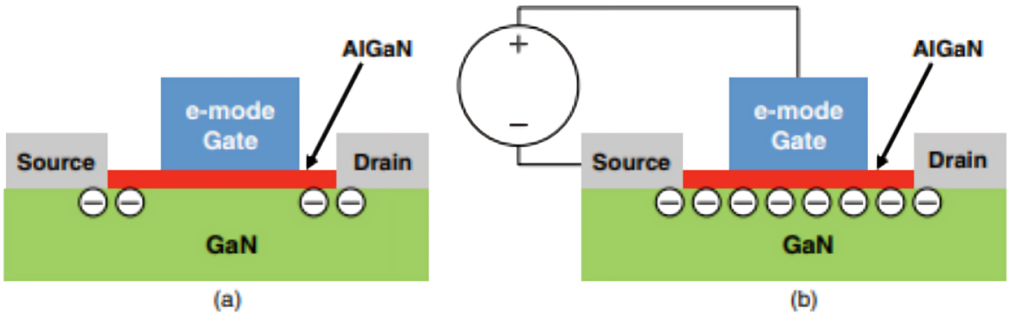


Figure 6: (a) An enhancement-mode (e-mode) device. (b) By applying a positive voltage to the gate, the electrons are attracted to the surface, re-establishing the 2DEG[37].

In order to have an Enhanced mode operation or normally-OFF operation, there are few possible solutions by Fabricating e-mode HEMT devices, they're as follows:

1. Using a recessed gate structure
2. A fluoride-based plasma treatment
3. Piezo neutralization technique

4. A p-GaN layer beneath the gate

5. Metal-oxide-semiconductor (MOS) structure

Possible solutions to make a normally ON HEMT device into a normally OFF HEMT device are depicted in Figure 7.

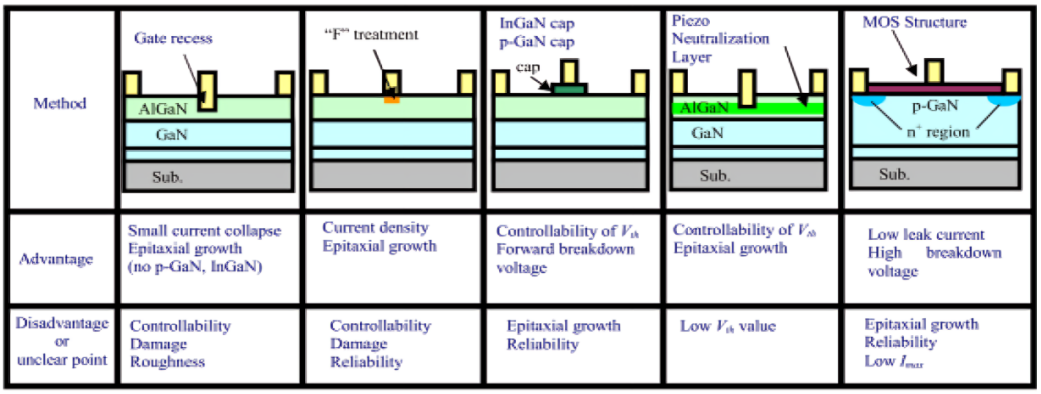


Figure 7: Approaches to realize enhancement mode GaN-based devices[38]

2.3 MOS HEMT

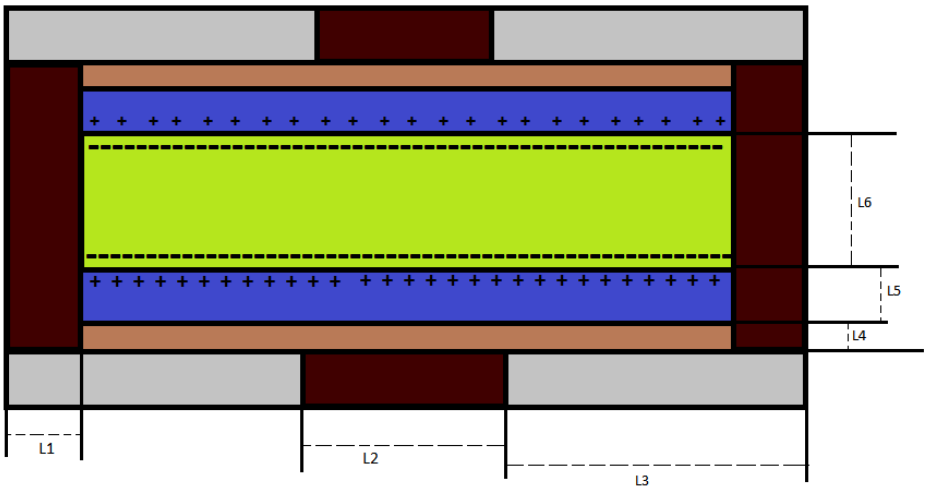


Figure 8: Cross section of a U-DG AlGaN/GaN MOS-HEMT with source and drain underlap[31]

A semiconductor device known as a MOS HEMT, or Metal-Oxide-Semiconductor High Electron-Mobility Transistor, combines the fundamentals of two different transistor technologies, namely MOSFET and HEMT. The high-performance transistor produced by this fusion has distinct properties that make it particularly well-suited for high-frequency, high power, and low-noise applications. A MOS HEMT's functioning is dependent on its heterojunction structure, which is made up of many semiconductor layers, each of which is essential to the operation of the device.

2.3.1 Operation Principle of MOS-HEMT

This device is a normally ON device or a Depletion mode device means Vth is a negative voltage i.e, at Vg=0 V device is in ON state and to make it OFF

we have to provide more negative voltage than Vth. When Vg>Vth, the device will be in ON State and by increasing Vg> 3V, Id starts decreasing, because the electrons will start flowing from the gate terminal, not from the drain terminal. Also the electric field will be very strong under the Gate Electrode than the entire channel, hence lead to the flow of electron directly from the Gate terminal.

Because of the polar GaN crystals, a substantial polarization effect is seen at the hetero junction interface [31]. GaN is chosen as the main device material because of its remarkable characteristics, including susceptibility to a range of temperatures and voltages. As a result, the device experiences a high electric field, which significantly improves the carrier concentration and spatial confinement of this quantum well. These AlGaN/GaN HEMT devices won't require doping because to the high 2DEG sheet carrier concentration of 1013 atoms/cm2, which is what makes them work. As a consequence, the non-uniform distribution of dopants and dopant dispersion are reduced. Due to the presence of III-V materials, the U-DG MOS-HEMT exhibits devices characteristics of both DG-MOSFETs and HEMTs, such as excellent drain current and reduced delay. It also has suppressed SCEs and lower gate leakage with sublime off-state control.

2.3.2 Advantages of MOS-HEMT

a) Low Power consumption

b) High Power Handling

c) Compatibility with CMOS integration

d) Wide Temperature Range Operation

e) Reduces self-heating effects

f) Scalability and Miniaturization

2.3.3 Applications of MOS-HEMT

a) Millimeter-Wave and Terahertz Electronics

b) Frequency Synthesizers

c) Wireless Power Amplifiers

d) Radio Astronomy

e) Medical Imaging

f) Aerospace and Satellite Communication

3 Development of Simulation Model

3.1 Silvaco ATLAS

ATLAS is a physically-based 2-D and 3-D device simulator that offers exceptional capabilities. It predicts the electrical behaviour of designated semiconductor structures and providesvaluable insight into the intricate internal physical mechanisms associated with device operation. By delving into the fundamental principles governing semiconductor devices,ATLAS serves as an indispensable tool for researchers, engineers, and developers seeking a deeper understanding of their designs and aiming to optimize device performance.

ATLAS produces three types of output files:

1. Run-Time Output
2. Log file
3. Solution file

The Figure 9[39] shows the type of information that moves into and out of the ATLAS.

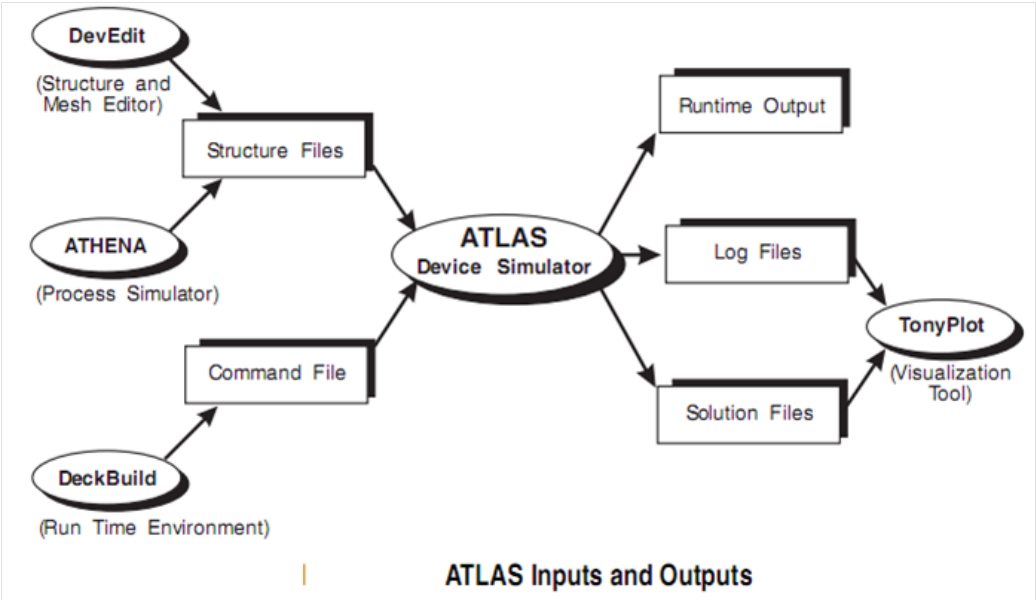


Figure 9: ATLAS Inputs and Outputs

3.2 The Order of ATLAS Commands

The order in which we should write the SILVACO Atlas commands is shown in Figure 10:

Group	Statement
1. Structure specification	Mesh Region Electrode Doping
2. Material models specification	Material Model Contact Interface
3. Numerical methods selection	Method
4. Solution specification	Log Solve Load Save
5. Result analysis	Extract Tony plot

Figure 10: ATLAS Command Groups with the Primary Statements in each Group[39].

3.2.1 Structure Specification

Running ATLAS inside Deckbuild

Every ATLAS run inside DECKBUILD should start with the line:

go atlas

An individual input file may contain multiple ATLAS runs, so each input file will be separated with a *go atlas* line.

Specifying The Initial Mesh

Syntax for defining Mesh points:

X.MESH LOCATION= <VALUE> SPACING=<VALUE>

Y.MESH LOCATION= <VALUE> SPACING=<VALUE>

Mesh covers the physical simulation domain. Mesh is constructed through used defined mesh points and spacing between them.

Meshing can be coarser or finer depending on the device simulation. ATLAS employs triangular meshes. Some triangulations yield better results than others.

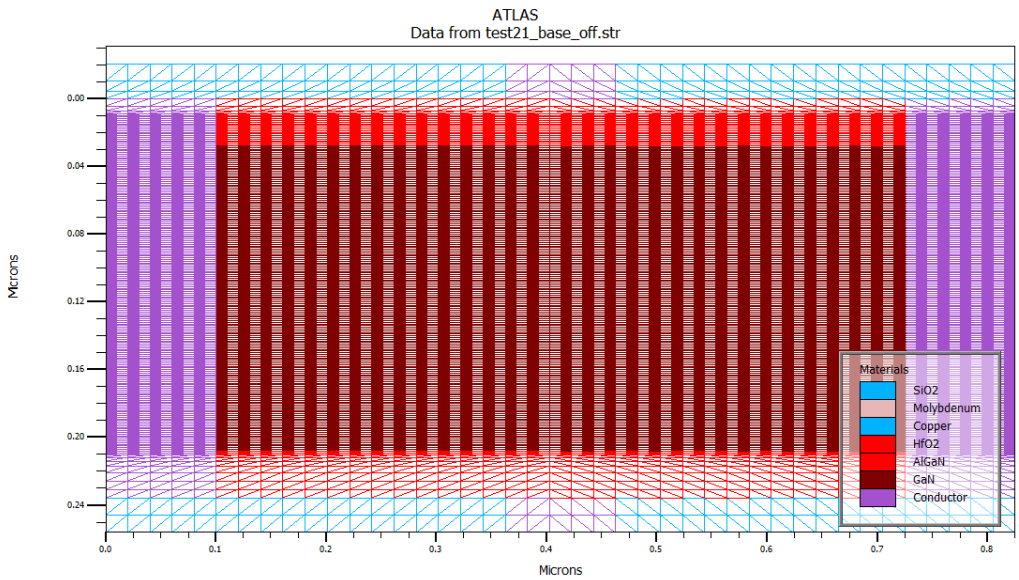


Figure 11: Meshing of DG DC MOS HEMT device(Calibrated)[31]

All the dimensions are mentioned in microns. For symmetrical devices, the mesh points must be symmetrical also, along x-axis and y-axis.

Specifying Regions and Materials

Syntax for defining Regions and Materials:

REGION number=<integer> material=<nameofMATERIAL> <PositionParam-

eters>

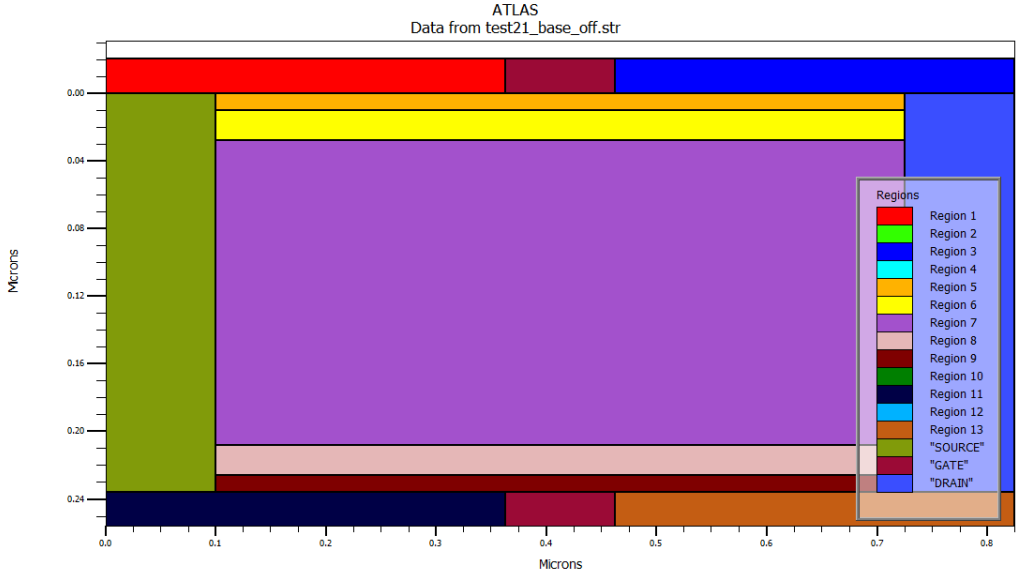


Figure 12: Regions of DG DC MOS HEMT device(Calibrated)[31]

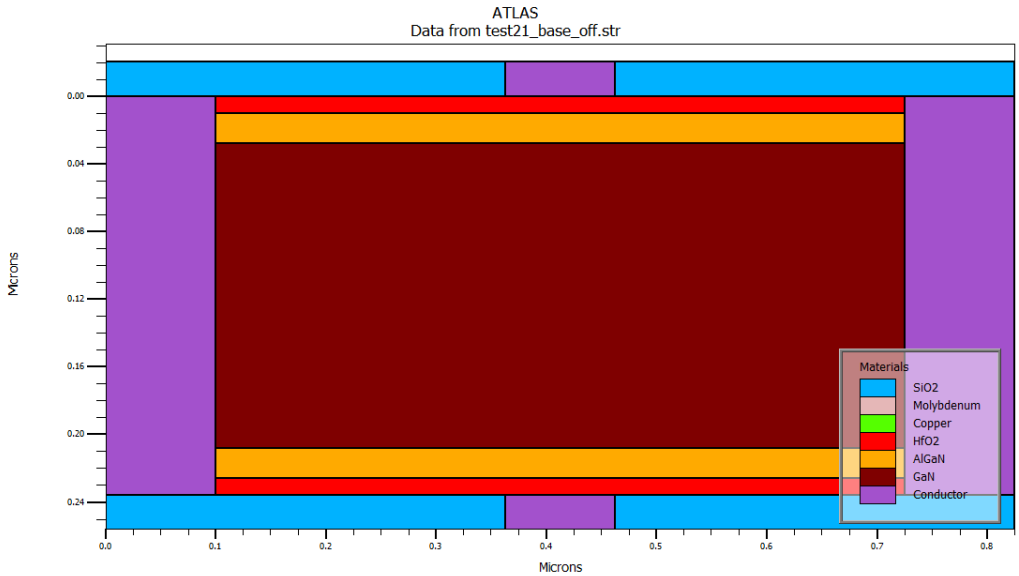


Figure 13: Materials corresponding to each region of DG DC MOS HEMT device(Calibrated)[31].

Specify the Material corresponding to specific region by varying one of the dimension and keeping other dimension as constant. Ex: special material in region 1 from x.min to x.max while keeping y position as constant.

Specifying Electrodes

Syntax for defining Electrodes:

ELECTRODE NAME=<nameofELECTRODE> <PositionParameters>

Specifying Doping

Syntax for defining Doping:

DOPING <DistributionType> <PositionParameters> <DopantType>

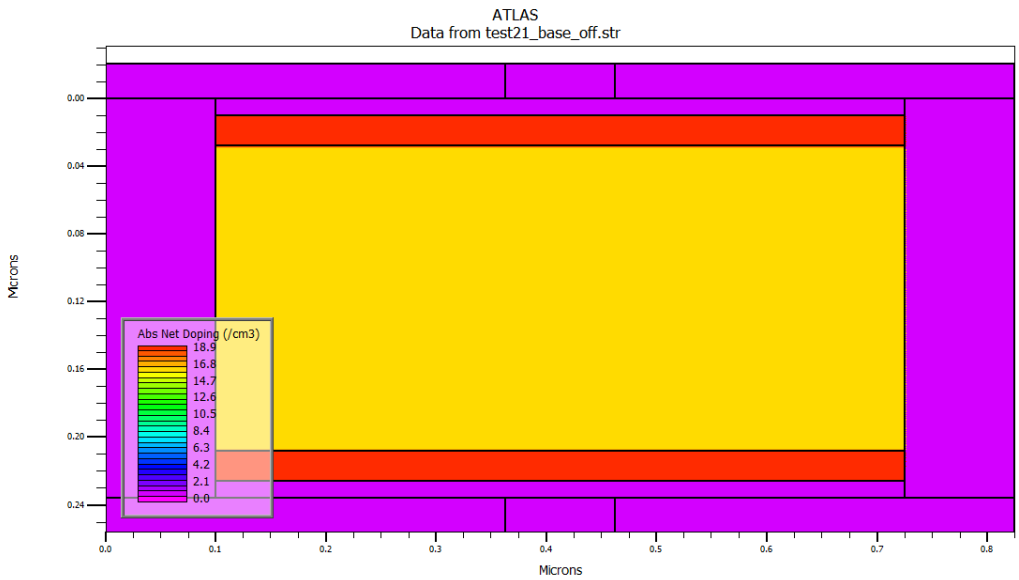


Figure 14: Doping density in different regions of DG DC MOS HEMT device (Calibrated)[31]

Minimum doping is required for the operation of the MOS-HEMT, resulting in increased mobility. Device is Uniformly doped for the proper

operation.

3.2.2 Material Model Specifications

Specifying Contact Characteristics: Work function for Gates or Schottky Contacts:

The CONTACT statement used to define the metal workfunction of one or more electrodes.

Syntax for defining Workfunction:

CONTACT NAME=<NameofELectrode> WORKFUNCTION=<ValueofWorkfunction>

Shorting Two Contacts Together If your designed structure is having multiple ELECTRODE statements with the same Functionality but defining separate locations within the device structure then, these electrodes must be Shorted so that same current will flow through both the ELECTRODES.

There are two methods of shorting two electrodes (located at different locations), Example: Double Gate MOS-HEMT:

Syntax for Shorting Two Electrodes:

a. Using COMMON keyword

ELECTRODE NAME=GATE 1 <positionalvalueGATE1>

CONTACT NAME=GATE2 COMMON=GATE1

b. Using same name of the Electrodes performing same functionality

ELECTRODE NAME=GATE <positionalvalue1GATE>

ELECTRODE NAME=GATE <positionalvalue2GATE>

Physical Models for Simulation of Device

Physical models are included using the MODELS and IMPACT statements. Parameters for these models may appear in many statements including: MODELS, IMPACT, MOBILITY, and MATERIAL [39]. Example: Models include in proposed work i.e, DG DC MOS-HEMT:

Syntax for including MODELS:

MODELS CVT SRH DGLOG AUGER KSN=0 PRINT

MOBILITY ALBRCT.N mod.watt.n

IMPACT N. CONCANNON

1. **Lombardi CVT Model:** The mobility model includes dependencies on the lattice temperature, channel doping level and normal electric field through Lombardi model[31].

2. **Density Gradient Model (DGLOG):** This model is included to predict the movement of Charge carriers i.e, both electrons and holes within the Semiconductor Material.Also during simulation DGLOG considers the spatial variation of charge carrier density across the device structure that is, it takes into account how the carrier concentration changes at different locations, which is influenced by the factors such as doping profiles, temperature, and bias conditions[39].

3. **Recombination Model Flag:** Recombination mechanism and carrier generation are incorporated using

   a. Auger recombination model (AUGER)

   b. Hydrodynamic Model (KSN=0)

c. Shockley Read-Hall (SRH) recombination model[31]

4. **Albrecht Model:** Included for Low Field Mobility in GaN[39] and to simulate low field mobility calibration in the device[40].

5. **Modified Watt Model:** Extension of original WAT Model to accurately describe the electron mobility in all regions of the Device. It takes into account various scattering mechanisms that affect electron mobility, such as phonon scattering, surface roughness scattering, and Coulomb scattering. This model is helpful for simulating latest scaled-down devices, where quantum effects and other non-classical phenomena are considered.

6. **Concannon's Impact Ionization Model:** This model is used to simulate impactionization, includes both electron and hole impact ionization coefficients by including N.CONCANNON and P.CONCANNON respectively. This model is useful for precisely predicting the breakdown behavior of semiconductor devices when they are subjected to high voltages or electric fields.

3.2.3 Numerical Method Selection

To obtain non-linear solution method, and associated parameters like iteration and convergence criteria will be mentioned in the METHOD statement using the NEWTON, GUMMEL, or BLOCK parameters. Combinations of these parameters can also be specified.[39]

Syntax for generating Non-Linear Solutions:

METHOD <solutiontechnique> trap=<integer> itlim=<integer> maxtrap=<integer>

Convergence Criteria: Non-linear iteration techniques converge at a rate, either linear or quadratic. The convergence is linear when Gummel iteration is used. The convergence is quadratic for Newton iteration. The non-linear iteration is terminated when the errors are acceptably small. The conditions required for termination of iteration are called Convergence Criteria.

3.2.4 Solution Specification

ATLAS is able to calculate DC, AC, small signal and transient solutions. We generally define voltages on each of the electrodes in the device. ATLAS then calculates the current through each of the electrode. ATLAS also calculates internal quantities for ex, carrier concentrations and electric field of the complete device.

Every simulation of the device initiates with zero bias on all the electrodes. Solutions are generated by stepping the biases on electrodes with reference to initial equilibrium condition. Always save results using the LOG or SAVE statements, and analyze and display these results using TONYPLOT.

To find DC Solutions, SOLVE Statement is used. Syntax of Solve Statement:

a. To supply direct DC value to a particular electrode:

SOLVE <nameofelectrode>=<real number>

b. To supply DC value in step wise manner, starting from initial value, incrementing by

Step size and ending to particular final value:

SOLVE <nameofelectrode> vstep=<stepsizevalue> name=<nameofElectrode> vfinal=<finalvalue>

Importance of Initial Guess

Convergence Problem occur near the beginning of the ramp when two previous results are not available. In order to obtain convergence for the equations used, supply a good initial guess for the variables to be evaluated at each bias point. The ATLAS solver uses initial assumptions and iterates to a converged solution. For isothermal drift diffusion simulations, the variables are the potential and the two carrier concentrations. In ATLAS simulations, convergence challenges often arise when employing a standard grid, primarily because of inadequate initial guesses for the solution. To address this issue, a careful approach is adopted during the biasing process, where the initial assumptions for each bias point are derived from a projection based on the two most recent generated results. By using this method to refine the initial guess, ATLAS endeavors to achieve better convergence and accuracy in its simulations, ensuring reliable predictions of semiconductor device behavior under various biasing conditions. If one last bias point is available, it will be used alone. This explains why the following two examples eventually produce the same result. The first will likely have far more convergence problems than the second.

To allow the ATLAS solver to find initial solutions, use the following syntax:

SOLVE init

3.2.5 Results Analysis

Run-Time Output: Run-time output is printed in the console Window of DeckBuild. To knowdetails of material parameters, constants and mobility models use the PRINT in the models statement and details will be specified

at the start of the run-time output.

Log Files: These files will store the current and voltages for each electrode in DC simulations or the terminal characteristics calculated by ATLAS solver.

Syntax to save and include the LOG File:

log outf=<filename>.log ————to open LOG File

logoff ————to close the LOG File

Tonyplot <filename>.log ———to view the LOG File in Tonyplot

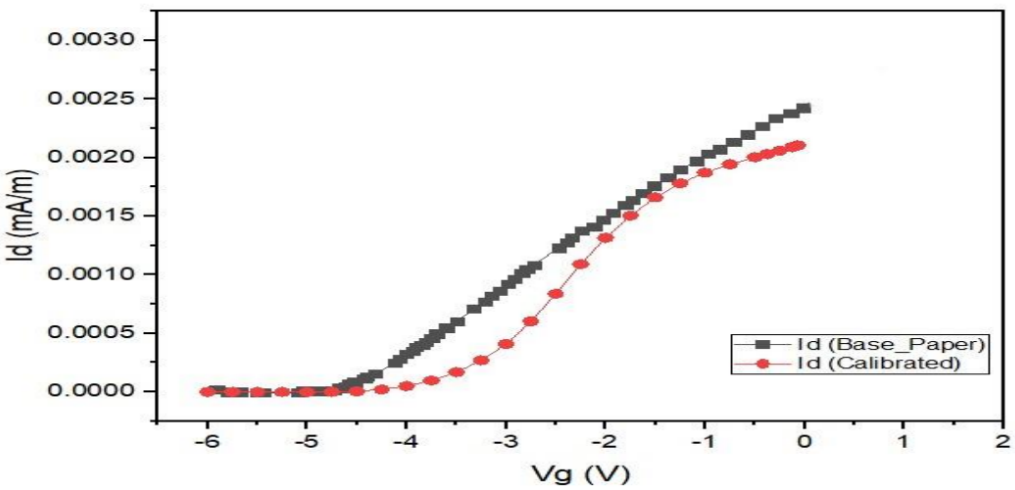


Figure 15: Comparison graph of variation of Id in linear scale as a function of Vgs at Vds = 2.0 V, for a U-DG AlGaN/GaN MOS-HEMT device of Lg=100nm[31].

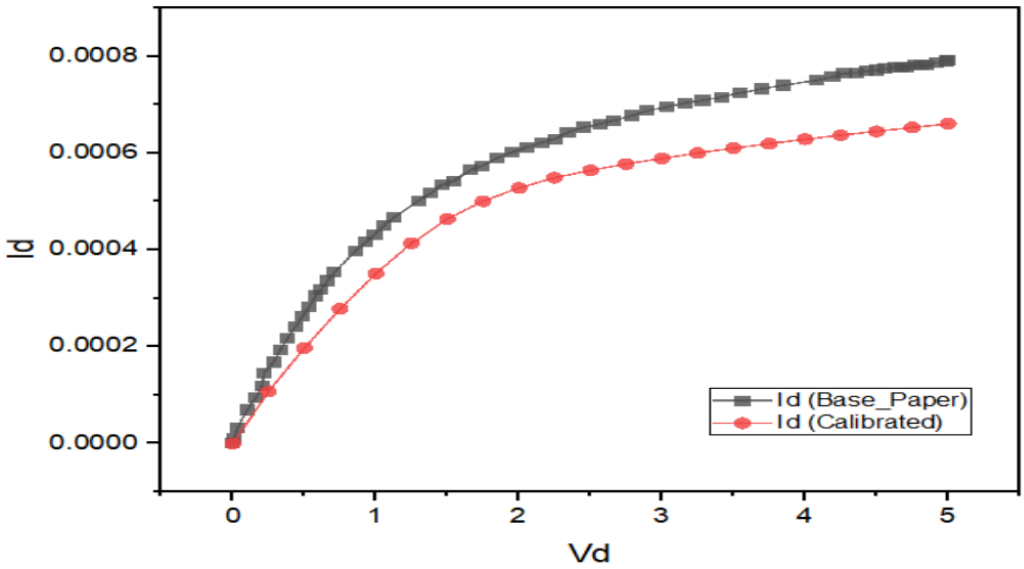


Figure 16: Comparison graph of variation of Id in linear scale as a function of Vds at VGS= 1.0V, for a U-DG AlGaN/GaN MOS-HEMT device of Channel length Lg=100nm[31].

4 Results and Discussions

In this chapter, the results from the device simulation are discussed here. The electrical performance of the device was performed by changing the desired parameters in the Silvaco TCAD Atlas file. The analysis is done by using Silvaco Tonyplot where the graph will be analyzed to get the desired parameters . The data of parameters taken from the Silvaco Tonyplot will be plot using Microsoft Excel Tools.

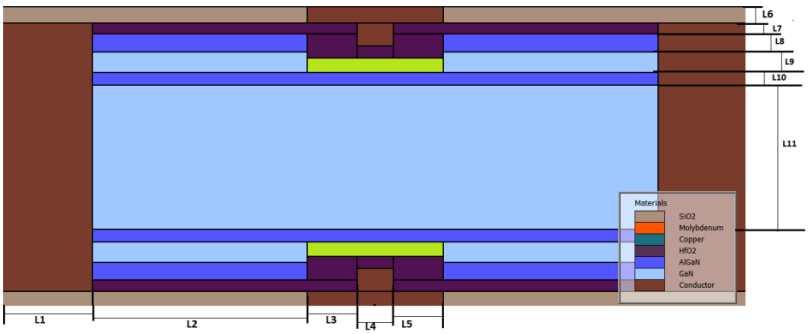
4.1 Results and Discussions

For improving the device performance of DG DC ALGaN/GaN based MOS-HEMT the structure proposed is:

4 Channel Double Recessed Gate AlGaN/GaN based MOS-HEMT for Power Amplifier Applications.

4.1.1 Proposed Structure: 4 Channel P-Doped Double Recessed Gate AlGaN/GaN based MOS-HEMT.

To make a normally ON MOS-HEMT device to a normally OFF MOS-HEMT device and simultaneously increasing the level of the current another technique proposed is 4 Channel Double Recessed Gate AlGaN/GaN based MOS-HEMT.



L1: 0.1µm
L2: 0.2375µm
L3: 0.055µm
L4: 0.04µm
L5: 0.055µm
L7: 0.02µm
L8: 0.014µm
L9: 0.022µm
L10: 0.016
L11: 0.18 µm

Figure 17: 4 Channel P-Doped Double Recessed Gate AlGaN/GaN based MOS-HEMT.

Case	Doping	Vth	Idss	Ion/Ioff	Gm_max	Cgs_max
1a	AlGaN: 9.e18 Ga	-0.883181 V	0.00349008	4.69565	0.00107	5.24E-15
1b	AlGaN: 9.e19 Ga	-4.48258 V	0.00421215	4.7005	0.00133	5.75E-15
1c	AlGaN: 9.e19 Ga	-0.919211 V	0.00422242	5.33109	0.001333	5.75E-15
1d	AlGaN: 9.e20 Ga	0.234545 V	0.00533826	1.280359	0.008424	2.05E-15
1e	AlGaN: 9.e18 Ga	Not Working				

Figure 18: Varying Doping Concentration of AlGaN/GaN and keeping Work function=5.0 and x composition=0.3 of AlGaN as constant.

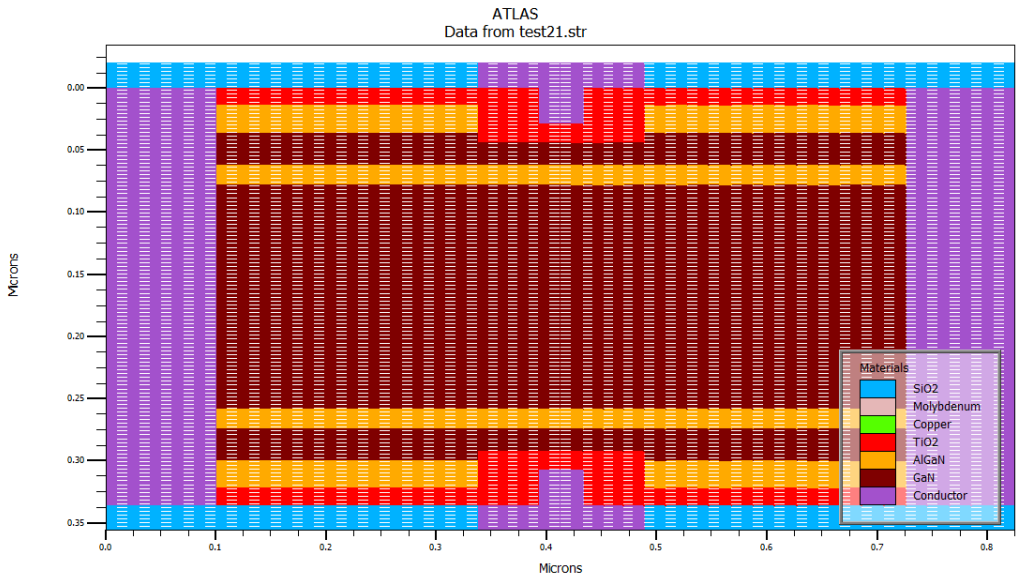


Figure 19: Mesing

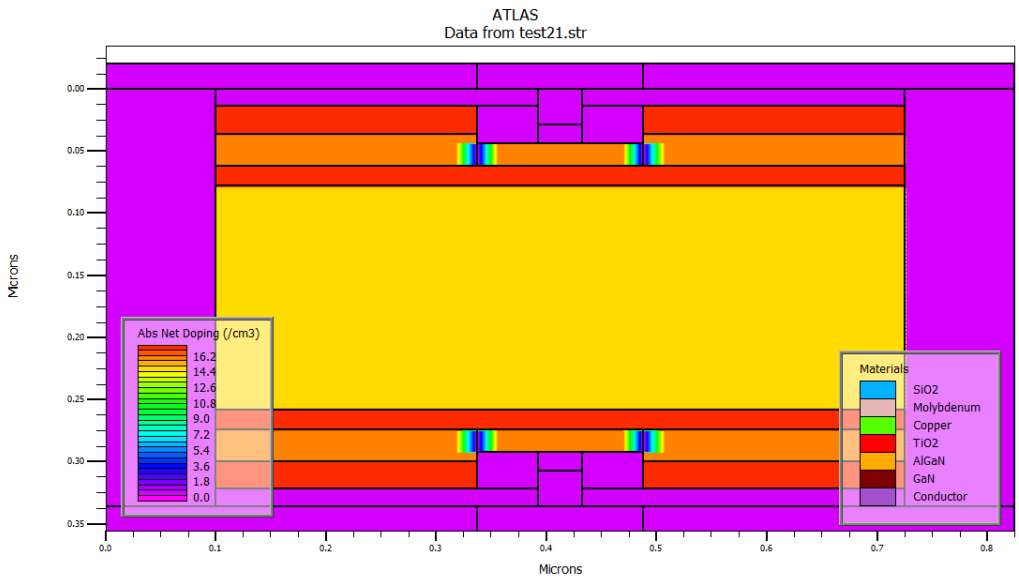


Figure 21: Doping Concentration Profile

When no of channels increased of the structure presented in Figure 8, the

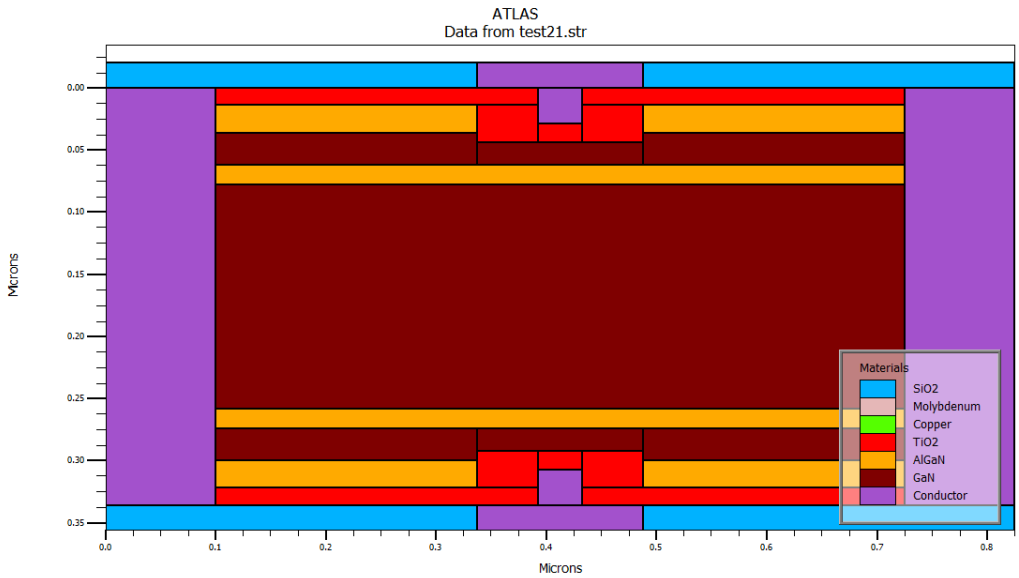


Figure 20: Materials Profile

level of current increased tremendously. To control the level of current various techniques have been employed like:

a) Increase the length of the Gate electrode for increasing the recession and,

b) Using p-GaN doping under the Gate electrode, shown in Figure [51].

The behavior of a MOS (Metal-Oxide-Semiconductor) High-Electron-Mobility Transistor (HEMT) can be affected by the introduction of p-GaN (p-type Gallium Nitride) under the gate electrode in the following ways, resulting in an increase in threshold voltage and a drop in current, because of the existence of the p-GaN layer can cause band bending in the channel area, also known as charge depletion. This band bending may result in a depletion region, narrowing the actual channel width and obstructing the flow

of charge carriers. A drop in current is caused by this depletion effect.

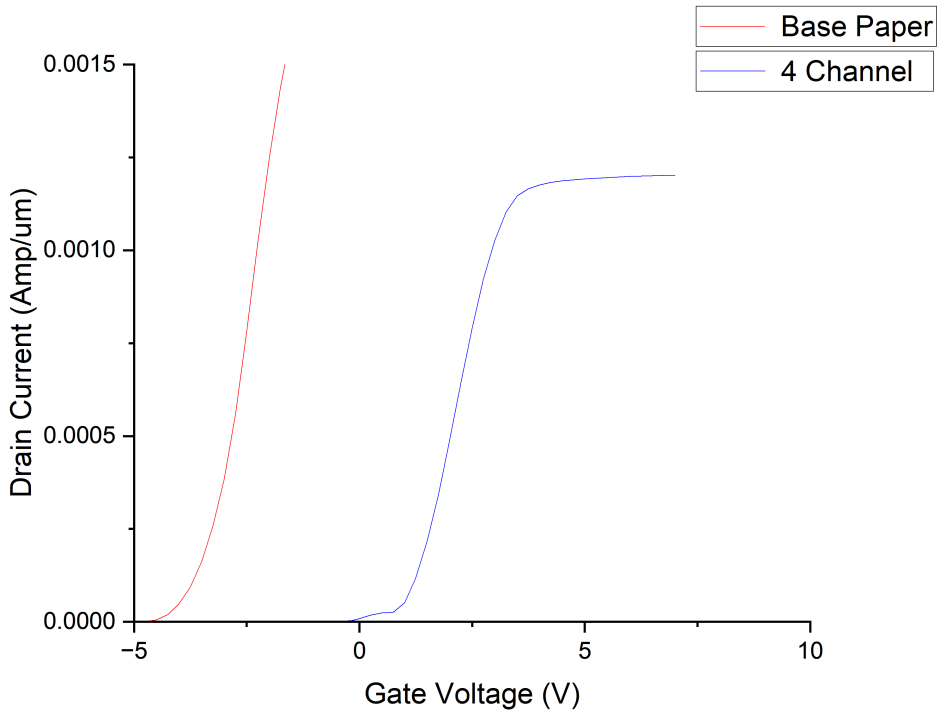


Figure 22: Transfer Characterisctics

The Vth of the proposed structure is increased and now the device is performing as normally OFF Device as the Vth=0.83379 V.

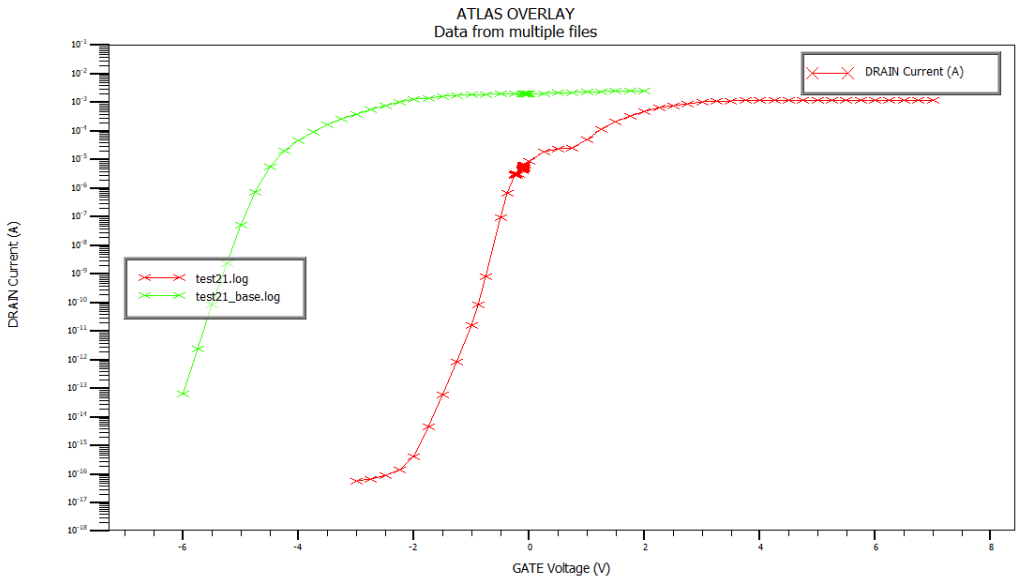


Figure 23: Transfer Characteristics in log scale

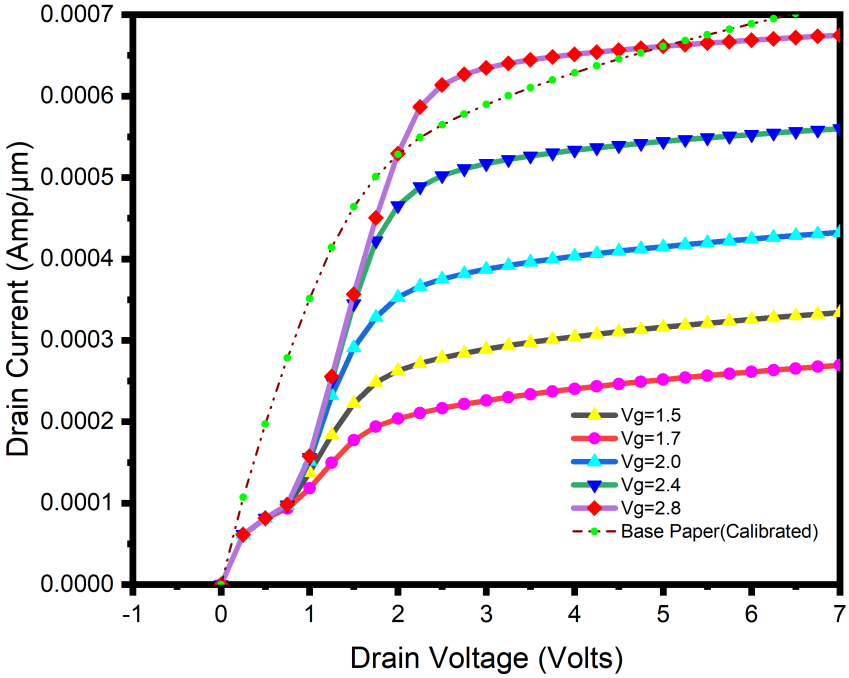


Figure 24: Output Characteristics comparison with various Gate Voltage

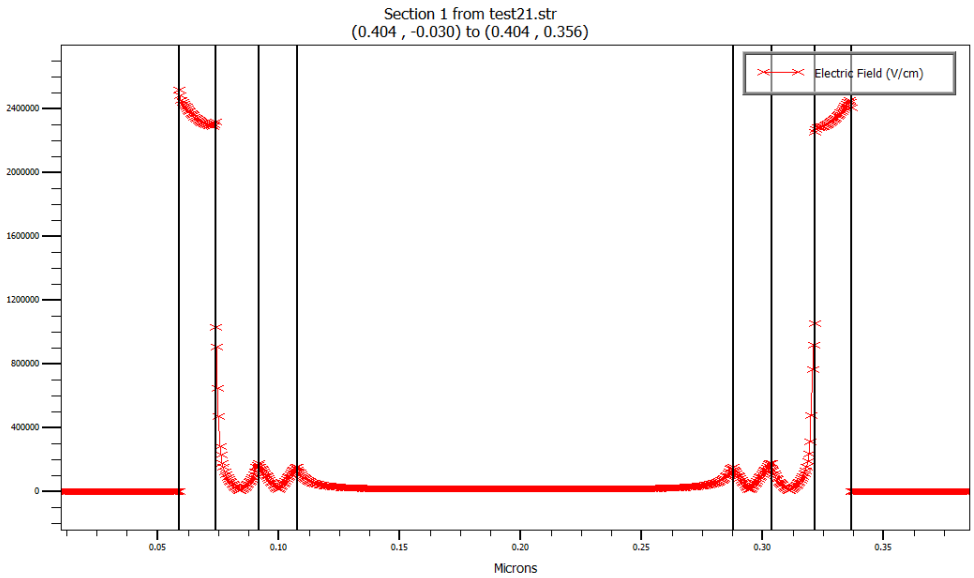


Figure 25: Electric Field Profile

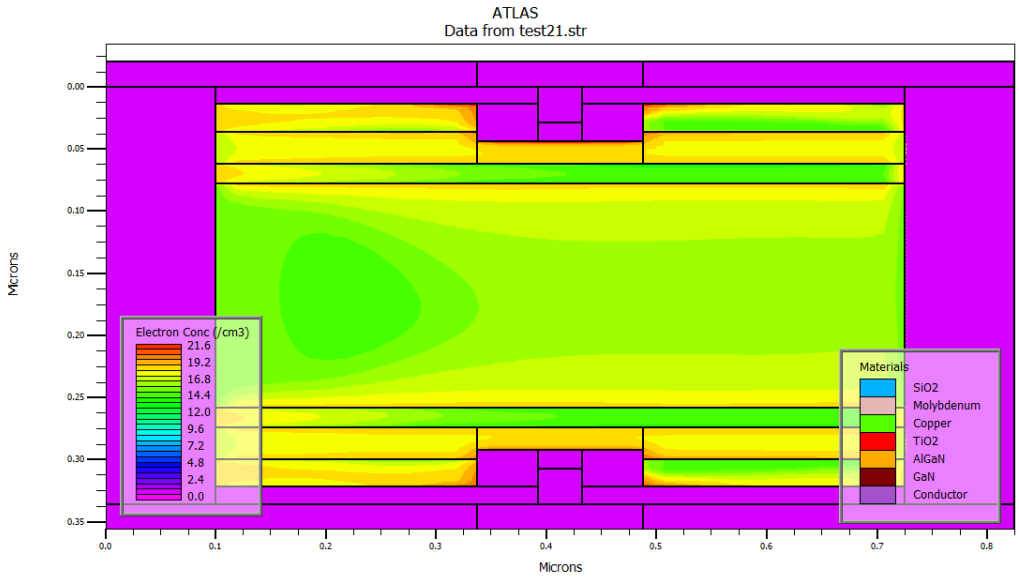


Figure 26: Electron Concentration Profile at 0V

Total Current Density Profile: Due to Recessed Gate electrode, the current

is negligible under the Gate Electrode.

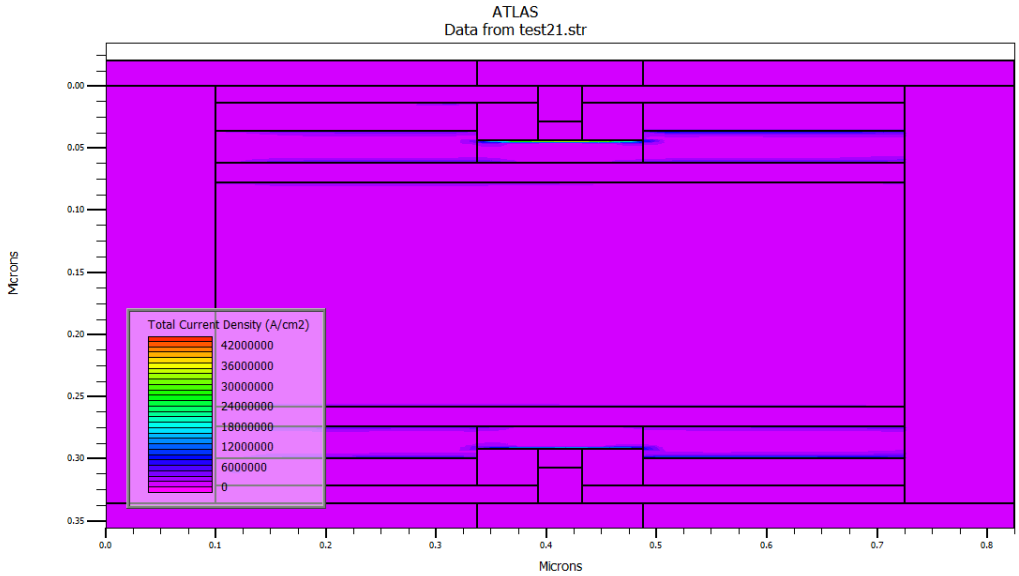


Figure 27: Current Density Profile

4.1.2 Summary

Various structure profile has been discussed and overall the structure presented in the previous section can be employed based on the application.

According to the objective of this dissertation is to make a Normally OFF device, therefore overall device presented, 4 Channel P-Doped Double Recessed Gate AlGaN/GaN based MOS-HEMT is better performing device among all because transfer characteristics, output characteristics, transconductance gm curve and the threshold voltage plots are better than other plots.

5 Conclusion and Future Scope

5.1 Conclusion and Future Scope

In Chapter 1, we have discussed the literature review which explains the need of a normally ON device and construction of MOS-HEMT device.

In Chapter 2, we have discussed about the basics of Material properties, working of HEMT and MOS-HEMT device. Also, the techniques are discussed for making a Normally ON device from Normally OFF device.

In Chapter 3, we calibrated MOS-HEMT structure which a normally ON device, which also explains only technique might not suffice for making a E-Mode device, various other device structure parameters are required to be altered.

In Chapter 4 we have discussed various experiments performed using SILVACO TCAD for making a normally OFF device, for example,by varying doping levels up to certain limit all these parameters are varied over 2 different structures. We have also discussed the comparison between different structures parameters like electron Concentration, Electric Field and Total Current Density.

5.2 Future Scope

Various other structures can be constructed like Triple Gate Controlled device which will increase controllability of the device. Alternate high bandgap material combinations in place of AlGaN/GaN can be employed for example, Al y In (1-x) N, where composition of x and y can be varied and high bandgap material can be achieved which will increase the depth

of 2DEG well and increase the height of the barrier. Another combination is GaN/InGaN [44] which will result in normally OFF device with Vth=+0.53. This device can be analyzed for AC analysis and various other parameters can be calculated based on which advantages of the device can be analyzed. The dimensions of the device can be reduced further so that power density [45] can be improved further, because nowadays, power density is an important parameter because to increase the performance of the device we require large amount current, high speed, high frequency, in smaller area so that to use the device, issue of area can be resolved. Also by minimizing the size of the device, energy, material, cost can be saved and overall affordable and efficient device can be manufactured.

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